A conditional Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$ from four arithmetic inputs

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Adversarial-audit revision

Abstract

We prove that four specific arithmetic statements in OpenAI's manuscript on Hilbert's tenth problem imply that the literal subset $\mathbb{Z} \subseteq \mathbb{Q}$ is Diophantine. These statements are taken as hypotheses; their proofs, including their companion-paper dependencies, are not independently verified here. The finite-certificate construction and its Diophantine integer model on a rank-one elliptic group are proved here in full. The new bridge defines a support relation between a native rational number and an integer label, with completeness on every nonzero integer, and a congruence relation whose precision is a variable integer label. Four labelled contact forms then bound the native rational height using the source's existing polynomial-in-radical estimate. A precision computed in the integer model separates the intended scalar from every other rational number within that height bound. The resulting formula has fixed finite length and uses neither a stronger height conjecture nor a prior scalar definition of the integers. A further internal adversarial audit found no unresolved gap in the conditional deduction presented here. This statement of audit outcome is not independent peer review or formal verification.

1 Statement and scope

Scope of the audit. The main conclusion is a positive answer to the native scalar-definability problem under the four arithmetic inputs explicitly listed below. The new bridge is Theorem 10.1; its proof starts in Section 10 and finishes in Section 14. Sections 2–9 prove the Diophantine integer model that the bridge uses; no earlier note is invoked as a lemma. The full deduction, including the integer-model construction, has been checked internally for denominator errors, fixed-formula uniformity, bad-prime witnesses, and circular use of scalar evaluation. No unresolved gap was identified in that deduction. The four external arithmetic statements are assumed, not re-proved or certified by this audit. In particular, the manuscript does not claim that an unconditional resolution has been independently verified. Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$, and heights are absolute logarithmic Weil heights with natural logarithms.

Theorem 1.1 (Conditional scalar-definability theorem)**.** *Assume the pole-parity proposition, the rank-one proposition, the five-point height theorem, and the prime-pattern lemma of [1], with the compatible fixed data in Section 2. Then there exist $N \geq 1$ and*

$$D(X, Y_1, \ldots, Y_N) \in \mathbb{Z}[X, Y_1, \ldots, Y_N]$$

such that

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^N \; D(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

More precisely, there is an existential definition of the graph $\{(n, nP) : n \in \mathbb{Z}\}$ in the fixed rational coordinate coding of the elliptic group.

The proof is the combination of Corollary 9.2, Theorem 10.1, and Corollary 14.3. No additional unproved height estimate or finite-fold representation hypothesis is used. The four named source results, rather than the abstract undecidability or Turing-completeness statement alone, are essential inputs.

For $R \in \{\mathbb{Z}, \mathbb{Q}\}$, let $\mathrm{H10}(R)$ denote the set of codes of polynomials with integer coefficients that have a zero in R, allowing the number of variables to vary. The distinction between a Turing reduction and a computable many-one reduction is essential here. The source manuscript explicitly obtains the former but does not claim the latter [1, Section 7]. We use specific arithmetic results from that manuscript, not the abstract assertion of Turing completeness alone.

Theorem 1.2 (Intermediate finite-certificate reduction)**.** *Assume the pole-parity proposition, the rank-one proposition, the five-point height theorem, and the prime-pattern lemma of [1], with the compatible fixed data stated in Section 2. Then there is a total computable map $f \mapsto Q_f$ from integral polynomials to integral polynomials such that*

$$(\exists \mathbf{n} \in \mathbb{Z}^r)\ f(\mathbf{n}) = 0 \quad \Longleftrightarrow \quad (\exists \mathbf{y} \in \mathbb{Q}^{N(f)})\ Q_f(\mathbf{y}) = 0.$$

Consequently $\mathrm{H10}(\mathbb{Q})$ *is Σ_1^0-complete under computable many-one reductions. The output polynomial can, after an effective further conversion, be taken to have total degree at most four.*

The point of the theorem is not an abstract collapse of Turing reducibility to many-one reducibility. The proof uses the particular pole-parity, height, elliptic, and prime-pattern inputs of the source manuscript. No Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$, finite-fold conjecture, or additional height conjecture is assumed.

2 Fixed arithmetic inputs

We record precisely the interfaces needed from [1], retaining its notation where convenient. All the data in this section are fixed independently of the input polynomial. In the source files, the four imported results have labels `par:pole-parity`, `ell:rank`, `hei:height`, and `pri:pattern`, respectively. Their proofs appear in Sections 3, 5, 4, and 6 of [1]. The formal-group estimates and simultaneous integer witnesses used below are also proved here from the fixed elliptic and prime-pattern data. The implications below require these four compatible arithmetic interfaces, not merely the conclusion that $\mathrm{H10}(\mathbb{Q})$ is undecidable or Turing-complete. All additional cited ingredients are established results of classical Diophantine definability, computability, and elliptic-curve theory.

Input P: pole parity. There is a positive existential ring formula $\Phi(t)$ with rational coefficients, and a finite set S_Φ of rational primes, such that

$$\mathbb{Q} \models \Phi(t) \qquad (t \in \mathbb{Z}), \tag{2.1}$$

$$\Phi(t), \quad q \notin S_\Phi, \quad v_q(t) < 0 \qquad \Longrightarrow \quad v_q(t) \in 2\mathbb{Z}. \tag{2.2}$$

This is the input proved in Section 3 of [1]. As usual, $v_q(0) = +\infty$, and valuations of nonzero rational numbers are normalized by $v_q(q) = 1$.

Input E: the rank-one group. Put

$$F = \mathbb{Q}(\sqrt{2}), \qquad E:\ y^2 = x^3 - d^2 x, \qquad d = 75 - 53\sqrt{2},$$

and let σ be the nontrivial automorphism of $F/\mathbb{Q}$. For a finite place w of F, write $\mathcal{O}_{F,w} = \{u \in F : w(u) \geq 0\}$ for the valuation ring on F, equivalently the localization of $\mathcal{O}_F$ at its associated prime ideal. There are a fixed positive integer m and a fixed nontorsion point $P \in E(F)$ such that

$$G := mE(F) = \mathbb{Z}P. \tag{2.3}$$

Use $z = -x/y$ on nonidentity points of G, and set $z(O) = 0$. Let ℓ_k be the degree-k truncation of the formal logarithm of E. Its coefficients are effectively computable from the fixed equation. The following formal-group estimates hold outside a fixed finite set of rational primes. They agree with Proposition `ell:formal` of [1]; a proof is given in Lemma 3.1. If $U = aP$ and $V = bP$, with $a, b \in \mathbb{Z}$ and $b \neq 0$, both reduce to O at a place w above q normalized by $w(q) = 1$, then

$$w\left(\frac{z(U)}{z(V)}\right) = v_q(a/b). \tag{2.4}$$

If additionally $q > k + 1$ and $v_q(a/b) \geq 0$, then

$$w\left(\frac{\ell_k(z(U))}{\ell_k(z(V))} - \frac{a}{b}\right) \geq k, \qquad \ell_k(z(V)) \neq 0. \tag{2.5}$$

The corresponding assertions hold for E^σ and P^σ. For a finite point with $x \neq 0$ on either of the good integral equations, reduction to O is equivalent to positive valuation of $1/x$.

Input H: the five-point height estimate. There are five distinct rational points $\mathcal{C} \subset \mathbf{P}^1(\mathbb{Q})$, including infinity, and fixed primitive integral linear forms $L_b(U, V)$ vanishing at $b \in \mathcal{C}$, with $L_\infty(U, V) = V$. For a suitable finite exceptional set S and fixed constants $H, c > 0$, the estimate in Section 4 of [1] is

$$h(s) \leq HM_S(s)^c \qquad (s \in \mathbf{P}^1(\mathbb{Q}) \setminus \mathcal{C}). \tag{2.6}$$

Here h is absolute logarithmic Weil height, and $M_S(u : v)$ is the product of the primes $q \notin S$ for which at least one normalized contact order

$$v_q(L_b(u, v)) - \min\{v_q(u), v_q(v)\} \tag{2.7}$$

is odd. We may enlarge c to a positive integer.

Choose a fixed finite set S_* containing S, S_Φ, 2, all fixed elliptic exceptions, the prime divisors of the determinants of pairs of contact forms, and the prime divisors of the multipliers in the next paragraph. Put $S_k = S_* \cup \{q : q \leq k + 1\}$ and $M_k = M_{S_k}$. There is a computable integer-valued function $k \mapsto H_k \geq 1$ such that

$$h(s) \leq H_k M_k(s)^c. \tag{2.8}$$

Indeed, choose an integer $H_0 \geq \max(1, H)$, put $D_* = \prod_{q \in S_*} q$, and take

$$H_k = H_0(D_*(k + 1)!)^c. \tag{2.9}$$

Enlarging the exceptional set in (2.6) costs at most the cth power of the product of the newly excluded primes.

Input R: compatible prime patterns. For the same five points and the same curve, there is a fixed finite set $\mathcal{K} \subset \mathbb{Z}_{>0}$ such that every integer A has a representation

$$Av = u_1 + u_2 + u_3, \qquad u_1, u_2, u_3, v \in \mathbb{Z}, \qquad v \neq 0, \tag{2.10}$$

in which every entry of the list

$$v, \qquad L_b(u_i, v) \quad (1 \le i \le 3,\ b \in \mathcal{C} \setminus \{\infty\}) \tag{2.11}$$

equals $\epsilon\kappa r$, with $\epsilon \in \{1, -1\}$, $\kappa \in \mathcal{K}$, and r a positive prime, as large as any prescribed lower bound. Each such prime splits in F and is good for both E and E^σ. At either place above r, let d_1, d_2 be the orders of the finite reduction groups of E, E^σ. Then

$$\gcd(d_1, d_2) \mid 4, \qquad r \nmid d_1 d_2. \tag{2.12}$$

This is precisely Lemma `pri:pattern` in Section 6 of [1]. The finite set $\mathcal{K}$ is independent both of A and of the lower bound on the primes. No assertion that a rational variable is prime will occur in the output formula. Primes are needed only to prove that actual integer solutions supply witnesses.

A fixed canonical-height lower bound. The usual canonical-height comparison [3], also used in Section 2 of [1], permits a fixed integer $a_* \ge 1$ with

$$h(x(tP)) \ge \frac{t^2}{a_*} - a_* \qquad (t \in \mathbb{Z} \setminus \{0\}). \tag{2.13}$$

All fixed rational data and all fixed integers above can be hardcoded in a single algorithm. Their existence is sufficient for the existence of that algorithm. This is an existence proof of a computable reduction; the note does not provide numerical values for all the fixed arithmetic constants. Importantly, (2.9) supplies the needed effective dependence on the input-dependent precision k.

3 Formal-group estimates and simultaneous integer witnesses

This section makes explicit the two elementary elliptic consequences used later. It does not require a Diophantine definition of an integer index.

Lemma 3.1 (Formal-parameter comparisons)**.** *After excluding a fixed finite set of rational primes, equations* (2.4) *and* (2.5) *hold with the hypotheses stated above, on both curves.*

Proof. Exclude 2, the ramified primes of $F/\mathbb{Q}$, and the primes of bad reduction of either integral equation. Let w be a remaining place above q, normalized by $w(q) = 1$. The completion is unramified, so its value group is $\mathbb{Z}$. For the parameter $z = -x/y$, the formal group is defined on the maximal ideal, and its invariant differential has the form

$$\left(1 + \sum_{j \ge 1} b_j Z^j\right) dZ$$

with integral coefficients at w. Consequently the coefficient of Z^t in its logarithm satisfies $w(c_t) \ge -v_q(t)$. These are the standard integral formal-group facts; see [3, Chapter IV]. For nonzero Z with $w(Z) \ge 1$, every term of degree $t \ge 2$ has valuation at least

$$t\, w(Z) - v_q(t) > w(Z),$$

and these lower bounds tend to infinity. The convergent logarithm is a homomorphism on the formal group and satisfies

$$w(\ell(Z)) = w(Z).$$

The logarithm identity at $Z = 0$ uses the infinite-valuation convention. If $U = aP$ and $V = bP$ are formal, with $b \neq 0$, the equality $[b]U = [a]V$ gives

$$b\ell(z(U)) = a\ell(z(V)).$$

The point V is nonidentity, so the logarithm of its parameter is nonzero. Taking valuations proves (2.4). The same statement includes $U = O$ with the infinite-valuation convention.

Suppose additionally that $q > k + 1$. For every integer $t > k$, one has

$$t - 1 - v_q(t) \geq k.$$

For $t < q$ the assertion is immediate. For $t \geq q$ with $j = v_q(t) > 0$, it follows from

$$t - 1 - j \geq q^j - 1 - j \geq q - 2 \geq k;$$

the case $v_q(t) = 0$ is again immediate. It follows that

$$w(\ell(Z) - \ell_k(Z)) \geq w(Z) + k \qquad (w(Z) \geq 1). \tag{3.1}$$

Put $\lambda = a/b$ and assume $v_q(\lambda) \geq 0$. If $a = 0$, the numerator comparison is exact, so only the nonvanishing of the denominator needs the tail estimate. For $a \neq 0$, proceed as follows. The valuation identity just proved gives $w(z(U)) \geq w(z(V))$. Subtracting the two logarithmic tails in $\ell(z(U)) = \lambda\ell(z(V))$ yields

$$w(\ell_k(z(U)) - \lambda\ell_k(z(V))) \geq w(z(V)) + k.$$

Equation (3.1) also gives $w(\ell_k(z(V))) = w(z(V))$, so this denominator is nonzero. Dividing proves (2.5). The argument is unchanged on the conjugate curve. □

The truncations are computable over F. For example, with $w_0 = -1/y$ the curve equation becomes

$$x = z/w_0, \qquad w_0 = z^3 - d^2 z w_0^2.$$

Solving recursively for w_0 as a series in z, substituting in $dx/(2y)$, and integrating term by term computes every coefficient required in ℓ_k. Only these finite polynomials, not infinite series, occur in our existential formulas.

Lemma 3.2 (Simultaneous integer witnesses)**.** *Fix $k \geq 1$ and a prime r supplied by the prime-pattern input, outside the fixed elliptic exceptions and with $r > k + 1$. At either chosen place $w \mid r$ there are nonzero integers h, j, independent of $a \in \mathbb{Z}$, such that all of $hP, haP, jP^\sigma, (h-4)aP^\sigma$ are formal at that place and*

$$w\left(\frac{\ell_k(z(haP))}{\ell_k(z(hP))} - a\right) \geq k, \tag{3.2}$$

$$w\left(\frac{z((h-4)aP^\sigma)}{z(jP^\sigma)}\right) \geq k. \tag{3.3}$$

Both denominators are nonzero.

Proof. Let d_1, d_2 be the two reduction orders in (2.12), and put $j = d_2$. The Chinese remainder theorem gives a positive integer h satisfying

$$h \equiv 0 \pmod{d_1}, \qquad h \equiv 4 \pmod{d_2 r^k}.$$

Indeed, $r \nmid d_1$ and $\gcd(d_1, d_2) \mid 4$ give exactly the compatibility condition. The reduction orders put hP, haP in the formal kernel of E and $jP^\sigma, (h-4)aP^\sigma$ in the formal kernel of E^σ.

The first comparison follows from Lemma 3.1 with index ratio a. For the second comparison that lemma gives

$$w\Big(\frac{z((h-4)aP^\sigma)}{z(d_2P^\sigma)}\Big) = v_r\Big(\frac{(h-4)a}{d_2}\Big) \geq k.$$

The denominator points are nonidentity, because their indices are nonzero and P is nontorsion. The nonvanishing of the truncated-logarithm denominator also follows from Lemma 3.1. This reproduces the simultaneous-witness consequence of the prime-pattern lemma without imposing any uniformity beyond the cited input. □

4 Finite guarded localization

The following elementary use of (2.2) is the new local ingredient.

Lemma 4.1 (Fourth-power guard)**.** *Suppose $q \notin S_\Phi$, $r \in \mathbb{Q}^\times$, and $v_q(r) = e$ is odd. If $\Phi(r)$ and $\Phi(rt^4)$ hold, then $e \geq 1$ and*

$$v_q(t) \geq -\lfloor e/4 \rfloor. \tag{4.1}$$

Proof. An odd negative valuation of r is excluded by $\Phi(r)$, so the odd integer e is positive. The assertion is immediate for $t = 0$. Otherwise $v_q(rt^4) = e + 4v_q(t)$ is odd and therefore cannot be negative. Rearranging this inequality, using the integrality of $v_q(t)$, gives (4.1). □

Write

$$\Gamma(r;t) := \Phi(rt^4).$$

For fixed rational variables r, b, C, impose

$$r \neq 0, \qquad \Phi(r), \qquad b^2 - 2 = rC, \qquad \Gamma(r;C). \tag{4.2}$$

These variables will be shared by all certificates attached to this occurrence of r. The rational parameter b here is a residue-branch witness; it is unrelated to the subscript indexing the fixed contact forms L_b.

Definition 4.2. For a positive integer k and $D \in F$, the formula $\mathrm{Loc}_{r,b,k}(D)$ has rational witnesses X_0, X_1, Y_0, Y_1, Z, W satisfying

$$X = X_0 + X_1\sqrt{2}, \qquad Y = Y_0 + Y_1\sqrt{2}, \qquad Y \neq 0,$$
$$DY = r^k X, \tag{4.3}$$
$$(Y_0 + bY_1)Z = 1 + rW, \tag{4.4}$$

and $\Gamma(r;t)$ for each of the six rational witnesses t. The root conditions (4.2) are imposed separately.

Every F-equation in this definition is a pair of rational equations in the basis $1, \sqrt{2}$. Thus this is an existential formula over $\mathbb{Q}$; it does not quantify over ideals or valuation rings.

Lemma 4.3 (Soundness of guarded localization)**.** *Assume (4.2), and let $q \notin S_\Phi \cup \{2\}$ have odd valuation $e = v_q(r)$. Then q splits in F, and there is a place $w \mid q$ with $\sqrt{2} \equiv b \pmod{w}$ such that, for every certificate attached to the same r, b, C,*

$$\mathrm{Loc}_{r,b,k}(D) \quad \Longrightarrow \quad w(D) \geq k. \tag{4.5}$$

Proof. Set $d_e = \lfloor e/4 \rfloor$. By Lemma 4.1, $e \geq 1$ and every guarded rational witness has valuation at least $-d_e$. In particular

$$v_q(b^2 - 2) = v_q(rC) \geq e - d_e > 0.$$

It follows that $v_q(b) = 0$, that 2 is a square modulo q, and that q splits in F. Choose the place w at which $\sqrt{2}$ reduces to b. Since $q \neq 2$, the factor $\sqrt{2} + b$ is a unit at this place, and hence

$$w(\sqrt{2} - b) = v_q(b^2 - 2) \geq e - d_e. \tag{4.6}$$

Here $b^2 - 2 \neq 0$ because b is rational.

For a certificate (4.3)–(4.4), the valuation of rW is at least $e - d_e > 0$. The right side of (4.4) is therefore a unit. Writing $Y_b = Y_0 + bY_1$ gives

$$v_q(Y_b) = -v_q(Z) \leq d_e. \tag{4.7}$$

On the other hand, (4.6) gives

$$w(Y - Y_b) = w\big(Y_1(\sqrt{2} - b)\big) \geq e - 2d_e > d_e.$$

The strict inequality follows from $e > 3\lfloor e/4 \rfloor$. Consequently $w(Y) = v_q(Y_b) \leq d_e$. We also have $w(X) \geq -d_e$, since $\sqrt{2}$ is a unit at w. Equation (4.3) therefore yields

$$w(D) \geq ke - 2d_e \geq k.$$

For the last inequality, note that $2\lfloor e/4 \rfloor \leq e - 1$ and $k \geq 1$. The case $D = 0$ satisfies the conclusion automatically. All certificates use the same place w, because that place was selected solely from q and the shared value b. □

Lemma 4.4 (Integer-witness completeness)**.** *Let r be an odd rational prime split in F, and let $w \mid r$ be a chosen place. Choose an integer b reducing to $\sqrt{2}$ at w and put $C = (b^2 - 2)/r \in \mathbb{Z}$. If $D \in r^k \mathcal{O}_{F,w}$, then* (4.2) *and* $\mathrm{Loc}_{r,b,k}(D)$ *have witnesses for which all guarded rational variables are integers.*

Proof. The ring of integers of F is $\mathcal{O}_F = \mathbb{Z}[\sqrt{2}]$. Since $D/r^k \in \mathcal{O}_{F,w}$, write $D/r^k = X/Y$ with $X, Y \in \mathcal{O}_F$ and $Y \notin \mathfrak{p}_w$. The integer $Y_0 + bY_1$ is nonzero modulo r, so choose integers Z, W satisfying (4.4). All displayed equations now hold with the six witness coordinates integral. Each argument rt^4 of Φ, as well as r itself, is an integer, so (2.1) supplies every required guard. The root equation has the integral witness C already chosen. □

Remark 4.5. Only the explicitly listed certificate witnesses are guarded. The rational witnesses internal to Φ, and rational inverse variables introduced when translating inequations, are not guarded and do not generate further requirements. Thus the definition is finite and has the integer-witness completeness just proved.

5 A finite elliptic comparison formula

For a point U on E or E^σ, let $\mathrm{For}_{r,b}(U)$ abbreviate

$$U = O \quad \text{or} \quad \big(U \neq O,\ x(U)y(U) \neq 0,\ \mathrm{Loc}_{r,b,1}(1/x(U))\big).$$

This is a finite existential formula in rational coordinates. The parameter $z(U)$ is interpreted as zero on the identity chart and as $-x(U)/y(U)$ on the other chart. All points that occur below lie in the torsion-free groups G or G^σ, so the nonzero-coordinate conditions cause no problem for nonidentity points.

Fix rational numbers $a_1, \ldots, a_n$ and points $T_1, \ldots, T_n \in G$. For notation put $a_0 = 1$ and $T_0 = P$.

Definition 5.1. The formula $\mathrm{Cmp}_k(r;\mathbf{a},\mathbf{T})$ asserts (4.2) for some rational b, C, and the existence of points

$$H, J, R_1, \dots, R_n \in G, \qquad J \neq O, \qquad \ell_k(z(H)) \neq 0.$$

Set $R_0 = H$ and $V_i = R_i - [4]T_i$ for $0 \le i \le n$. For all these finitely many indices impose

$$\mathrm{For}_{r,b}(H), \quad \mathrm{For}_{r,b}(R_i), \quad \mathrm{For}_{r,b}(J^\sigma), \quad \mathrm{For}_{r,b}(V_i^\sigma), \tag{5.1}$$

$$\mathrm{Loc}_{r,b,k}\left(\frac{\ell_k(z(R_i))}{\ell_k(z(H))} - a_i\right), \tag{5.2}$$

$$\mathrm{Loc}_{r,b,k}\left(\frac{z(V_i^\sigma)}{z(J^\sigma)}\right). \tag{5.3}$$

All localization certificates share r, b, C but have their own six witnesses.

Membership in $G = mE(F)$ is existential: quantify a point $U \in E(F)$ and impose $[m]U = T$. The finite elliptic group-law relations, their conjugates, and their nonidentity conditions are expressible on finitely many algebraic charts. Consequently Definition 5.1 is effective and existential over $\mathbb{Q}$. It uses no integer-index variable in its syntax.

Lemma 5.2. *Suppose* $\mathrm{Cmp}_k(r;\mathbf{a},\mathbf{T})$ *holds and write* $T_i = n_i P$, *where* $n_i \in \mathbb{Z}$ *and* $n_0 = 1$. *For every* $q \notin S_k$ *such that* $v_q(r)$ *is odd, one has*

$$v_q(a_i - n_i) \ge k \qquad (1 \le i \le n). \tag{5.4}$$

In particular every a_i *is* q*-integral.*

Proof. Lemma 4.3 gives one place $w \mid q$ at which every certificate is sound. The formal conditions put the indicated points in the actual formal kernels at this place. Write

$$R_i = t_i P, \qquad H = t_0 P, \qquad J = jP, \qquad t_i, j \in \mathbb{Z}, \quad j \neq 0.$$

The index of V_i is $t_i - 4n_i$. Apply (2.4) on the conjugate curve to (5.3). It gives

$$v_q\left(\frac{t_i - 4n_i}{j}\right) \ge k.$$

Since j is an integer, $v_q(j) \ge 0$. Thus $t_i \equiv 4n_i \pmod{q^k}$ for all $0 \le i \le n$. At $i = 0$ this says $t_0 \equiv 4 \pmod{q^k}$. Since $q \neq 2$, t_0 is a q-unit and

$$t_i/t_0 \in \mathbb{Z}_q, \qquad t_i/t_0 \equiv n_i \pmod{q^k}.$$

Now (2.5) applies to R_i, H because $q > k+1$ and their index ratio is integral. Together with the soundness of (5.2), it gives $a_i \equiv n_i \pmod{q^k}$. This congruence also proves the asserted integrality of a_i. □

Lemma 5.3. *For a prime* r *supplied by the prime-pattern input, chosen sufficiently large for precision* k, *and any integers* $a_i = n_i$ *with* $T_i = n_i P$, *the formula* $\mathrm{Cmp}_k(r;\mathbf{a},\mathbf{T})$ *holds.*

Proof. Choose either place $w \mid r$ and apply Lemma 3.2 to obtain its integers h, j. Take $H = hP$, $R_i = hn_iP$, and $J = jP$. Then $V_i = (h-4)n_iP$, including $V_0 = (h-4)P$. All required local inequalities and formal-kernel statements follow from Lemma 3.2. Choose one integer lift b of the residue of $\sqrt{2}$ at w. Lemma 4.4 supplies all localization certificates, including those for reciprocal x-coordinates, with this same b. Identity points use their identity charts. All guarded values are integers, so all pole-parity conditions hold. □

6 Finite contact representations

Definition 6.1. The formula $\mathrm{Rep}_k(A;\mathbf{a},\mathbf{T})$ asserts the existence of rational u_1,u_2,u_3,v satisfying $v\neq 0$ and $Av=u_1+u_2+u_3$. Every entry of (2.11) is required to equal $\epsilon\kappa r$ for a choice $\epsilon\in\{1,-1\}$, $\kappa\in\mathcal{K}$, and a nonzero rational r, and $\mathrm{Cmp}_k(r;\mathbf{a},\mathbf{T})$ is required for that occurrence. The rational variables and all auxiliary witnesses attached to different occurrences are independent, except for the common parameters $\mathbf{a},\mathbf{T}$. The choices of signs and multipliers are finite disjunctions.

An entry shared by different slopes may be supplied with a single occurrence or with repeated independent certificates. Either convention is finite and has the properties below.

Lemma 6.2. *If $A,a_1,\dots,a_n$ are integers and $T_i=a_iP$, then $\mathrm{Rep}_k(A;\mathbf{a},\mathbf{T})$ holds.*

Proof. Choose a representation from the prime-pattern input with all its selected primes larger than every fixed exception and larger than $k+1$. Use Lemma 5.3 at each factor. □

Lemma 6.3. *Let $f\in\mathbb{Z}[X_1,\dots,X_n]$ be nonconstant, set*

$$D=\deg f,\qquad C_f=\max\left(1,\sum_{\alpha}|c_\alpha|\right),\qquad k=cD,$$

where c_α are the coefficients of f. Suppose $f(\mathbf{a})=0$, $T_i=n_iP\in G$, and put

$$B=\max(1,|n_1|,\dots,|n_n|),\qquad \delta=f(n_1,\dots,n_n).$$

If $\delta\neq 0$ and $\mathrm{Rep}_k(A;\mathbf{a},\mathbf{T})$ holds, then

$$h(A)\le 3H_kC_fB+2. \tag{6.1}$$

Proof. Consider a slope $s_i=(u_i:v)$ in the representation, and a prime q counted by $M_k(s_i)$. All its contact forms are nonzero because $\epsilon\kappa r\neq 0$, so $s_i\notin\mathcal{C}$ and the cited height estimate applies. Outside S_*, the minimum of their valuations is $\min\{v_q(u_i),v_q(v)\}$. To verify this for arbitrary rational, not necessarily primitive, coordinates, use integrality of the form coefficients for one inequality and invert a pair of forms with q-unit determinant for the other. Since one normalized contact order is odd, the valuations of the raw contact forms cannot all have the same parity. One raw contact form therefore has odd valuation. Its expression $\epsilon\kappa r$, with κ a q-unit, gives an occurrence of r with odd $v_q(r)$. Lemma 5.2 applies to the comparison attached to this occurrence and gives

$$a_j\in\mathbb{Z}_q,\qquad a_j\equiv n_j\pmod{q^k}\quad(1\le j\le n).$$

Since f has integer coefficients and $f(\mathbf{a})=0$, it follows that $v_q(\delta)\ge k$. This holds for every prime in the squarefree product $M_k(s_i)$, so

$$M_k(s_i)^k\mid\delta.$$

The ordinary polynomial estimate gives $|\delta|\le C_fB^D$. By (2.8) and $k=cD$,

$$h(s_i)\le H_k|\delta|^{c/k}\le H_kC_f^{1/D}B\le H_kC_fB.$$

Finally $A=u_1/v+u_2/v+u_3/v$ and the height inequality for a sum gives

$$h(A)\le\sum_{i=1}^{3}h(s_i)+\log 3\le 3H_kC_fB+2.$$

□

7 Construction of the single finite formula

Let $f \in \mathbb{Z}[X_1, \ldots, X_n]$ be nonconstant, and keep D, C_f, k as in Lemma 6.3. Define the computable integer

$$N_f = 9a_* H_k C_f + a_*^2 + 8a_*. \tag{7.1}$$

In particular $N_f \geq 1$.

Construct an existential formula Ψ_f over $\mathbb{Q}$ as follows. It has rational root variables $a_1, \ldots, a_n$ and elliptic points $T_1, \ldots, T_n \in G$. Require

$$f(a_1, \ldots, a_n) = 0. \tag{7.2}$$

For each j, require either $a_j = 0$ and $T_j = O$, or $a_j \neq 0$ and $T_j \neq O$. In the latter case, introduce rational numbers $A_{j,0}, A_{j,1}, A_{j,2}$ with $A_{j,2} \neq 0$, and impose

$$x(T_j) = \frac{A_{j,0} + A_{j,1}\sqrt{2}}{A_{j,2}}, \tag{7.3}$$

together with

$$\mathrm{Rep}_k(A_{j,t}; \mathbf{a}, \mathbf{T}) \qquad (0 \leq t \leq 2). \tag{7.4}$$

The finite-point chart also imposes $x(T_j)y(T_j) \neq 0$. Finally, for every $1 \leq j \leq n$ and every ordinary integer s with $|s| \leq N_f$, impose

$$T_j \neq [s]P \quad \text{or} \quad a_j = s. \tag{7.5}$$

The finite list of exact points $[s]P$ is computable from the fixed point P, using additions and inversions over the computable field F. Equality and inequality with these points have the explicit encodings in Appendix A.

The phrase "for every" in (7.4) and (7.5) denotes the external construction of a finite conjunction, not a universal quantifier in the formula. No prime, height, or integer-index predicate appears in Ψ_f. All its quantified variables range over $\mathbb{Q}$, with F-valued coordinates represented by rational pairs.

Proposition 7.1. *For every nonconstant integral polynomial f,*

$$\mathbb{Q} \models \Psi_f \quad \Longleftrightarrow \quad (\exists \mathbf{n} \in \mathbb{Z}^n)\ f(\mathbf{n}) = 0.$$

Proof. First suppose $\mathbf{n}$ is an integer zero. Take $a_j = n_j$ and $T_j = n_j P$. For $n_j \neq 0$, the coordinates of T_j belong to F, and $x(T_j)$ admits (7.3) with all three $A_{j,t}$ integral. Lemma 6.2 supplies (7.4). If $T_j = [s]P$, the nontorsion property of P gives $n_j = s$, so (7.5) holds. The nonzero-coordinate conditions hold because nonzero points in G are nontorsion, whereas a point with $x = 0$ or $y = 0$ on the displayed equation is torsion. Thus Ψ_f has rational witnesses.

Conversely suppose Ψ_f has rational witnesses. By (2.3), write $T_j = n_j P$ with unique ordinary integers n_j. Set $B = \max(1, |n_1|, \ldots, |n_n|)$ and $\delta = f(\mathbf{n})$. If $\delta = 0$, we already have the desired integer zero. Assume therefore that $\delta \neq 0$.

For each nonidentity T_j, Lemma 6.3 bounds each of its three rational coordinate representatives by

$$h(A_{j,t}) \leq 3H_k C_f B + 2.$$

Using (7.3), the absolute height inequalities over F, and $h(\sqrt{2}) + \log 2 \leq 2$, we obtain

$$h(x(T_j)) \leq \sum_{t=0}^{2} h(A_{j,t}) + h(\sqrt{2}) + \log 2 \leq 9H_k C_f B + 8. \tag{7.6}$$

If $B > 1$, choose j with $|n_j| = B$. This point is nonidentity, and (2.13) and (7.6) give

$$\frac{B^2}{a_*} - a_* \le 9H_kC_fB + 8.$$

Consequently

$$B^2 \le 9a_*H_kC_fB + a_*^2 + 8a_* \le (9a_*H_kC_f + a_*^2 + 8a_*)B,$$

where the last inequality uses $B \ge 1$. Thus $B \le N_f$. The same conclusion is immediate when $B = 1$.

For each j, the condition (7.5) with $s = n_j$ now forces $a_j = n_j$. Equation (7.2) therefore gives $\delta = f(\mathbf{n}) = f(\mathbf{a}) = 0$, a contradiction. Hence the case $\delta \neq 0$ is impossible, and the integer zero exists. □

8 Effective polynomial compilation and consequences

All the constructions in Ψ_f are effective once the fixed arithmetic data are chosen. The order of choices matters: first fix the curve, m, P, the contact forms and multiplier list, Φ, the finite exceptional sets, and the integers c, H_0, a_*. Only then read f, compute its degree and coefficient sum, compute k, H_k, N_f, and write out the finite formula. No prime-pattern witnesses, integer indices, or rational-solvability answers are used by this compilation algorithm. More formally, every finite tuple of rational and integer literals of the required shape determines a computable compiler; the cited arithmetic results assert the existence of one tuple for which the correctness proof applies. Existence of that single correct compiler is exactly what the many-one reducibility assertion requires. Here are the relevant details to distinguish computable compilation from an oracle procedure. Appendix A supplies an explicit finite encoding of all the point predicates, so no negation of an existential membership predicate is being used.

An element of F is represented by a pair of rationals in the basis $1, \sqrt{2}$. Its arithmetic and conjugation are effective, and a nonzero element has nonzero norm in $\mathbb{Q}$. Elliptic membership, the fixed multiplication map $[m]$, finite additions, equality with $[s]P$, and their negations can be expressed on finitely many algebraic charts. Rational functions in these charts are used only together with their nonzero-denominator conditions. The truncation ℓ_k is a finite computable polynomial. Each occurrence of Φ is a fixed existential formula with a polynomial expression substituted for its parameter. Signs, multipliers, identity charts, and finite nonidentity conditions contribute only finite Boolean combinations.

Replace a rational inequation $g \neq 0$ by $gw = 1$ with a fresh rational variable. Move existential quantifiers outward and put the resulting finite formula into disjunctive normal form. For each conjunction of equations $g_1 = \cdots = g_t = 0$, use the single equation $g_1^2 + \cdots + g_t^2 = 0$. A finite disjunction of resulting equations is a single product equation, with separate auxiliary variables available for every branch. Finally clear the fixed rational coefficient denominators. This gives an integral polynomial Q_f with

$$\mathbb{Q} \models \Psi_f \quad \Longleftrightarrow \quad Q_f \text{ has a rational zero.}$$

For the constant zero input use the constant zero output; for a nonzero constant input use the constant one output. Thus the construction extends to a total computable map on all integral polynomials, proving the reduction in Theorem 1.2.

By the Davis–Putnam–Robinson–Matiyasevich theorem [2, Theorem 8.1], integer solvability is many-one complete among computably enumerable sets. Rational solvability is computably enumerable by enumeration and exact evaluation on rational tuples. The reduction therefore proves Σ_1^0-completeness of H10($\mathbb{Q}$), conditional precisely on the stated arithmetic inputs.

For the quartic assertion, compute an arithmetic circuit for Q_f and attach one variable to each gate. Its gate equations have degree at most two, and an output-zero equation imposes $Q_f = 0$. The sum of the squares of these gate equations has degree at most four and has a rational zero exactly when Q_f does. This is the usual effective circuit normalization, also described in Section 7 of [1].

Remark 8.1 (Limitation of the intermediate reduction)**.** This intermediate construction alone does not produce one existential formula defining $\mathbb{Z}$ in $\mathbb{Q}$. The number of comparisons with $[s]P$ in (7.5) depends on the computed cutoff N_f and can grow without bound with the input. A computable family of polynomials with input-dependent arity is not a single polynomial family with the input represented by a rational parameter. In particular, an arbitrary computable map from input codes to polynomials is not an existentially defined family in which those codes themselves range over rational parameters. No such stronger uniform representation follows merely from the many-one reduction. Sections 9–14 obtain it instead by retaining the point parameters, building arithmetic on their labels, and then constructing a separate scalar bridge.

Remark 8.2 (Where the infinite tests have been removed)**.** The guarded localization certificates replace both the all-ring pole-parity closure and the quotient-field localization arguments by explicit finite conditions. The contact formulas are required only for the three rational representatives of each relevant elliptic x-coordinate. The final bounded-index identification is enforced by the finite list (7.5), rather than by an additive injection defined on an entire model of a ring theory. No compactness theorem or search for a failed rational test remains in the reduction.

Remark 8.3 (What the cutoff does and does not bound)**.** The bound $B \le N_f$ was deduced only under the temporary assumption $f(n_1, \dots, n_n) \neq 0$. It is not a bound on the size of integer zeros of f. For a genuine integer zero, arbitrarily large indices are allowed: every implication in (7.5) is automatically true when $a_j = n_j$. This distinction is essential, since some input polynomials have integer zeros of unbounded size.

9 Retaining the elliptic parameters: a Diophantine integer model

The preceding argument proves more than the equivalence of two sentences. The elliptic points can be retained as free parameters. This distinction is important: the rational root variables need not equal their elliptic indices.

Theorem 9.1 (Fiberwise representation)**.** *For each fixed polynomial $f \in \mathbb{Z}[X_1, \dots, X_r]$, there is an effectively constructed existential formula $\Theta_f(T_1, \dots, T_r)$ over $\mathbb{Q}$ such that*

$$\mathbb{Q} \models \Theta_f(n_1P, \dots, n_rP) \quad \Longleftrightarrow \quad f(n_1, \dots, n_r) = 0 \qquad (\mathbf{n} \in \mathbb{Z}^r).$$

Point parameters are represented by their rational coordinate codes from Appendix A. All occurrences of G in the formula are existentially defined.

Proof. For nonconstant f, use the formula constructed in Section 7 but leave the point variables $T_1, \dots, T_r$ free. Quantify the rational variables a_j and all other witnesses as before. The forward direction of Proposition 7.1 supplies witnesses for each specified integer zero $\mathbf{n}$ with the prescribed points n_jP. Conversely, fix the free points at $T_j = n_jP$ and suppose the formula holds. The contradiction argument in that proposition proves $f(\mathbf{n}) = 0$ for these very indices, not merely the existence of some other integer zero. It does not assert that every auxiliary rational tuple $\mathbf{a}$ in a satisfying assignment equals $\mathbf{n}$. Only the relation on the

prescribed point indices is used to define multiplication. For a constant polynomial, use the identically true or identically false formula on G^r, respectively. In both cases the free point parameters are explicitly required to belong to G. □

Corollary 9.2 (Diophantine model)**.** *The map*

$$\iota : \mathbb{Z} \longrightarrow G, \qquad n \longmapsto nP,$$

is a Diophantine model of the ring of integers in a finite Cartesian power of $\mathbb{Q}$*. In particular, the relation*

$$\mathrm{Mul}_G(T, U, V) \quad \Longleftrightarrow \quad T = nP,\ U = mP,\ V = nmP$$

is existentially definable over $\mathbb{Q}$*.*

Proof. The set of point codes for G and the elliptic group law are existentially definable by Appendix A. The fixed point P represents 1 and the identity represents 0. Use Theorem 9.1 with the single fixed polynomial $X_1X_2 - X_3$ to define multiplication. The map ι is a computable bijection onto the chosen canonical codes for G. □

Corollary 9.3 (Computably enumerable relations on the labels)**.** *If* $A \subseteq \mathbb{Z}^r$ *is computably enumerable, then*

$$\widehat{A} := \{(n_1P, \ldots, n_rP) : (n_1, \ldots, n_r) \in A\}$$

is Diophantine over $\mathbb{Q}$*.*

Proof. The Davis–Putnam–Robinson–Matiyasevich theorem gives an integer polynomial $U(\mathbf{n}, \mathbf{w})$ whose integer solvability in $\mathbf{w}$ represents A; see [2, Theorem 8.1]. For clarity, its formulation for nonnegative or positive integers gives this signed-integer version without any additional hypothesis. Pull A back under $(\mathbf{u}, \mathbf{v}) \mapsto \mathbf{u} - \mathbf{v}$ on nonnegative integer tuples, apply the theorem there, impose $n_i = u_i - v_i$, and express nonnegativity of every restricted integer variable by a sum of four squares. Combining these finitely many integer equations by a sum of squares produces the stated polynomial U. Apply Theorem 9.1 to this fixed polynomial, retain the point parameters representing $\mathbf{n}$, and existentially quantify the point parameters representing $\mathbf{w}$. Equivalently, interpret its finite arithmetic circuit in the model of Corollary 9.2. □

10 The scalar bridge and its dependencies

We now pass from arithmetic on the labels nP to the actual scalar $n \in \mathbb{Q}$. The new argument uses the Diophantine model of Corollary 9.2, the pole-parity predicate, and the same five-point height estimate already stated in Section 2. It does not assume a stronger height estimate, a finite-fold representation theorem, or a Diophantine definition of a variable rational prime. The prime-pattern input has already been used to construct the model; the new scalar bridge does not use it again. The strategy of combining bounds with sufficiently strong congruences is motivated by Poonen's rank-one construction over rings of integers, especially the separation argument and formal-group approximation in [5, Lemmas 5 and 11]. Here a new support certificate supplies a bound for a native rational input in terms of an integer label, and arithmetic on the labels makes the required precision variable inside one formula.

Theorem 10.1 (Model-to-scalar bridge)**.** *Assume the Diophantine model on* $G = \mathbb{Z}P$ *from Corollary 9.2, the predicate* (2.1)–(2.2)*, and the five-point height estimate of Section 2. There is a single existential ring formula* $\mathrm{Eval}(x, T)$ *over* $\mathbb{Q}$ *such that*

$$\mathbb{Q} \models \mathrm{Eval}(x, nP) \quad \Longleftrightarrow \quad x = n \qquad (x \in \mathbb{Q},\ n \in \mathbb{Z}).$$

The point argument has its fixed finite rational coordinate code. Consequently the literal subset $\mathbb{Z} \subseteq \mathbb{Q}$ *is Diophantine under these hypotheses.*

The proof has two interfaces between native scalars and integer labels. The first is a relation $\mathrm{Supp}(r, nP)$ that constrains every odd-valuation prime of r to divide n, outside a fixed finite set, and that holds whenever $r = n \neq 0$. The second is an exact congruence relation with a variable precision label:

$$\mathrm{Cong}_3(x, nP, kP) \quad \Longleftrightarrow \quad v_3(x - n) \geq k \qquad (k \geq 0).$$

The first interface gives a logarithmic height bound computed entirely from the integer label. The second then separates the intended integer from every other rational number below that bound.

The fixed exceptional set. Enlarge a fixed finite set S so that it contains S_Φ, 2, 3, every rational prime below a bad-reduction place of the displayed integral equation for E, every ramified prime of $F/\mathbb{Q}$, and the exceptional primes in the five-point height estimate. Write the four finite contact forms as

$$L_i(U, V) = a_i U + b_i V, \qquad a_i, b_i \in \mathbb{Z}, \quad a_i \neq 0, \qquad 1 \leq i \leq 4, \tag{10.1}$$

and take $L_\infty(U, V) = V$. Include in S all prime divisors of the nonzero integers a_i and all fixed determinant exceptions for these forms. By the enlargement rule for the height theorem, there are fixed positive integers H, c such that

$$h(s) \leq H M_S(s)^c \qquad (s \in \mathbf{P}^1(\mathbb{Q}) \setminus \mathcal{C}). \tag{10.2}$$

Increasing the original positive real constants to suitable integers is harmless because $M_S(s) \geq 1$. All these choices are made once, not separately for each input x or each label nP.

11 A rational guarded localization certificate

The localization certificate needed for the new argument is simpler than the quadratic-field certificate in Definition 4.2. It applies directly to a rational number and has no choice of residue branch.

Definition 11.1. For $r, D \in \mathbb{Q}$, let $\mathrm{RL}(r, D)$ be the following existential condition:

$$r \neq 0, \qquad \Phi(r),$$

there exist $X, Y, Z, W \in \mathbb{Q}$ such that

$$Y \neq 0, \qquad DY = rX, \qquad YZ = 1 + rW, \tag{11.1}$$

and

$$\Phi(rX^4), \quad \Phi(rY^4), \quad \Phi(rZ^4), \quad \Phi(rW^4) \tag{11.2}$$

hold. Only these four coefficient witnesses are guarded. The internal witnesses in the fixed formula Φ are not recursively guarded.

Lemma 11.2 (Soundness)**.** *Suppose* $\mathrm{RL}(r, D)$ *holds,* $q \notin S_\Phi$*, and* $v_q(r)$ *is odd. Then* $v_q(r) > 0$ *and* $v_q(D) \geq 1$*, with the convention* $v_q(0) = +\infty$*.*

Proof. Set $e = v_q(r)$. The predicate $\Phi(r)$ excludes a negative odd value, so $e \geq 1$. Put $d_e = \lfloor e/4 \rfloor$. For each guarded witness t, the integer $e + 4v_q(t)$, when finite, is odd. The condition $\Phi(rt^4)$ therefore forces $e + 4v_q(t) \geq 0$, hence $v_q(t) \geq -d_e$. This conclusion also holds for $t = 0$. In particular, $v_q(rW) \geq e - d_e > 0$, so $YZ = 1 + rW$ is a q-adic unit. Thus $Z \neq 0$ and

$$v_q(Y) = -v_q(Z) \leq d_e.$$

If $D = 0$, the assertion is immediate. Otherwise $X \neq 0$ and the other equation gives

$$v_q(D) = e + v_q(X) - v_q(Y) \geq e - 2d_e \geq 1.$$

The last inequality holds for every positive odd integer e. □

Lemma 11.3 (Integer-modulus completeness)**.** *Let $m \in \mathbb{Z} \setminus \{0\}$ and $D \in \mathbb{Q}$. Suppose that*

$$v_p(D) \geq v_p(m) \qquad \text{for every rational prime } p \mid m. \tag{11.3}$$

Then $\mathrm{RL}(m, D)$ *has witnesses $X, Y, Z, W \in \mathbb{Z}$. For $m = \pm 1$, condition* (11.3) *is empty and the assertion holds for every $D \in \mathbb{Q}$.*

Proof. Write $D/m = X/Y$ with $X, Y \in \mathbb{Z}$, $Y \neq 0$, and $\gcd(Y, m) = 1$. Such a representation exists by (11.3); for $D = 0$ take $X = 0, Y = 1$. Choose $Z \in \mathbb{Z}$ with $YZ \equiv 1 \pmod{m}$, and put $W = (YZ - 1)/m \in \mathbb{Z}$. These choices also work for negative m; for $m = \pm 1$ one can take $Z = 0$. Equations (11.1) hold with $r = m$. Every argument of Φ in Definition 11.1 is now an ordinary integer, so all the required predicates hold by (2.1). □

12 A Diophantine support relation with diagonal completeness

For $J \in G \setminus \{O\}$, the coordinates $x(J)$ and $y(J)$ are nonzero. Indeed, on $y^2 = x^3 - d^2x$, a point with $x = 0$ or $y = 0$ is a point of order two, whereas G is torsion-free. Thus $1/x(J)$ and $z(J) = -x(J)/y(J)$ are defined and $z(J) \neq 0$.

Definition 12.1. For $r \in \mathbb{Q}$ and a point code T, define $\mathrm{Supp}(r, T)$ by requiring $r \neq 0$, $T \in G \setminus \{O\}$, and the existence of $J, R \in G \setminus \{O\}$ with

$$\mathrm{Mul}_G(T, J, R).$$

Introduce $a, b, D \in \mathbb{Q}$ by the rational-function identities

$$\frac{1}{x(J)} = a + b\sqrt{2}, \qquad D = N_{F/\mathbb{Q}}\left(\frac{z(R)}{z(J)}\right), \tag{12.1}$$

and require

$$\mathrm{RL}(r, a), \qquad \mathrm{RL}(r, b), \qquad \mathrm{RL}(r, D). \tag{12.2}$$

All rational functions are encoded with nonzero denominators, as in Appendix A. This is a fixed finite existential formula over $\mathbb{Q}$.

In the semantics of the model, if $T = mP$ and $J = jP$, the multiplication condition says $R = mjP = [m]J$. It does not multiply a native rational variable by a point using an undefined variable-length group operation.

Proposition 12.2 (Support control)**.** *Let $m \in \mathbb{Z} \setminus \{0\}$, and suppose* $\mathrm{Supp}(r, mP)$ *holds. For every prime $q \notin S$,*

$$v_q(r) \text{ odd} \quad \Longrightarrow \quad v_q(r) > 0 \ \text{ and } \ q \mid m. \tag{12.3}$$

Proof. Fix such a q with odd $v_q(r)$, and choose the witnesses in Definition 12.1. Lemma 11.2 gives

$$v_q(a) \geq 1, \qquad v_q(b) \geq 1, \qquad v_q(D) \geq 1.$$

At every place w of F above q, normalized by $w(q) = 1$, the first two inequalities imply

$$w(1/x(J)) = w(a + b\sqrt{2}) \geq 1.$$

The displayed integral equation for E has good reduction there, so $w(x(J)) < 0$ implies that J belongs to its formal kernel. The point $R = [m]J$ belongs to that kernel as well. The formal-logarithm valuation identity, proved in Lemma 3.1, gives

$$w\left(\frac{z(R)}{z(J)}\right) = v_q(m).$$

The identity holds at *every* place above q, not at a single arbitrarily chosen branch. Since q is unramified in F and $[F : \mathbb{Q}] = 2$, the norm satisfies

$$v_q(D) = \sum_{w|q} [F_w : \mathbb{Q}_q]\, w\left(\frac{z(R)}{z(J)}\right) = 2v_q(m).$$

Together with $v_q(D) \geq 1$, this implies $v_q(m) \geq 1$. The positivity of the odd valuation of r is part of Lemma 11.2. □

The reverse direction needed below is stronger than completeness for prime moduli. It includes arbitrary nonzero integers, with arbitrary powers of bad primes.

Proposition 12.3 (Diagonal completeness)**.** *For every $m \in \mathbb{Z} \setminus \{0\}$,*

$$\mathbb{Q} \models \mathrm{Supp}(m, mP). \tag{12.4}$$

Proof. First suppose $|m| > 1$. At every place w above every prime $p \mid m$, choose a sufficiently small formal subgroup about O for the fixed integral Weierstrass equation. The choice can be made so small that the formal logarithm is an isometry on it and so that, for any point J in all the chosen neighborhoods, the rational coefficients a, b of $1/x(J)$ satisfy

$$v_p(a), v_p(b) \geq v_p(m) \qquad (p \mid m). \tag{12.5}$$

Here the coordinates are interpreted simultaneously in $F \otimes_{\mathbb{Q}} \mathbb{Q}_p$. To justify the second requirement explicitly, $1/x(J)$ tends to zero at every place over p as J tends to O, and its rational coefficients are

$$a = \frac{1/x(J) + (1/x(J))^\sigma}{2}, \qquad b = \frac{1/x(J) - (1/x(J))^\sigma}{2\sqrt{2}}.$$

These are continuous $\mathbb{Q}_p$-linear projections. The fixed denominators at 2 merely require a deeper neighborhood. At a split prime these projections are applied to the two components of $F \otimes_{\mathbb{Q}} \mathbb{Q}_p$ simultaneously. Closeness in only one component would not suffice.

The argument uses formal groups also at bad or ramified primes, but only arbitrarily close to O. The formal group of an integral Weierstrass equation and its invariant differential have integral coefficients even when the special fiber is singular away from O. For the displayed short equation, this is especially transparent: with $v = -1/y$ one has $v = z^3 - d^2 z v^2$, and

$$\frac{dx}{2y} = \frac{dz}{\sqrt{1 + 4d^2 z^4}} = \sum_{j \geq 0} (-1)^j \binom{2j}{j} d^{2j} z^{4j}\, dz.$$

The square root here denotes a formal power series with constant term one, not an additional field extension or a quantified operation. Thus the invariant differential has integral coefficients at every finite place, and termwise integration gives the coefficient bound used next. If $w(p) = 1$, the formal logarithm is an isometry whenever $w(z) > 1/(p-1)$: its coefficient of degree t has valuation at least $-v_p(t)$, and

$$(t-1)w(z) > v_p(t) \qquad (t \geq 2).$$

On a sufficiently deep formal subgroup it is a homomorphism, is injective, and gives

$$w\left(\frac{z([m]J)}{z(J)}\right) = v_p(m). \tag{12.6}$$

This is the standard local formal-group argument; see [3, Chapter IV, §6]. No good-reduction hypothesis at the primes dividing m is needed for this deep-neighborhood assertion.

There is a nonzero integer j for which $J = jP$ lies in all these chosen subgroups. Indeed, each is an open subgroup of the compact local elliptic group and therefore has finite index. An integer multiple of P lies in each subgroup, and a common multiple works simultaneously at the finitely many places. This is the same compact-local-group mechanism used to force denominator divisibility in [5, Lemmas 6 and 10]. The resulting global point J is nonidentity because P is nontorsion. Put $R = [m]J$, which is also nonidentity.

For D in (12.1), equation (12.6) and the local norm formula give

$$v_p(D) = 2v_p(m) \qquad (p \mid m). \tag{12.7}$$

The formula remains valid at ramified places when valuations are normalized by $w(p) = 1$, since $\sum_{w|p}[F_w : \mathbb{Q}_p] = 2$. Indeed, if ord_w has value group $\mathbb{Z}$, then $\mathrm{ord}_w = e_w w$ and the norm formula is

$$v_p(N_{F/\mathbb{Q}}(u)) = \sum_{w|p} f_w \,\mathrm{ord}_w(u) = \sum_{w|p} e_w f_w w(u).$$

Substituting (12.6) gives $2v_p(m)$ independently of ramification. Thus a, b, D all satisfy the hypothesis of Lemma 11.3 with modulus m. Apply that lemma separately to the three quantities to obtain the integer coefficient witnesses for (12.2). The model multiplication relation holds because $R = [m]J$.

For $m = \pm 1$, choose any $J \in G \setminus \{O\}$ and let $R = [m]J$. Lemma 11.3 has no local hypothesis for these two moduli, so it again supplies all three certificates. This completes the proof for positive and negative nonzero integers. □

Remark 12.4 (What the support relation does not assert)**.** The relation $\mathrm{Supp}(r, mP)$ does not by itself imply $r = m$ or even $r \in \mathbb{Z}$. It constrains the prime support of the square class of r and has the diagonal completeness property (12.4). Those two properties, rather than an unproved exact prime or scalar decoder, are precisely what the height argument below uses.

13 Variable precision at one fixed prime

We next construct a fixed formula whose precision is an integer label, not the length of a list of equations. Only the usual existential definition of the fixed ring $\mathbb{Z}_{(3)}$ is needed from local definability.

13.1 The fixed local ring

For completeness, put

$$S_{a,b} = \{2t_1 : \exists t_2, t_3, t_4 \in \mathbb{Q},\ t_1^2 - at_2^2 - bt_3^2 + abt_4^2 = 1\}, \qquad T_{a,b} = S_{a,b} + S_{a,b}.$$

Koenigsmann's quaternionic description gives

$$\mathcal{R}_3 := T_{-1,3} + T_{-1,21} = \mathbb{Z}_{(3)}. \tag{13.1}$$

Here is the ramification check for this particular choice. For odd p and positive odd b, the Hilbert symbol is $(-1, b)_p = \left(\frac{-1}{p}\right)^{v_p(b)}$, and $(-1, b)_2 = (-1)^{(b-1)/2}$. Thus $(-1, 3)$ is ramified

exactly at $2, 3$, while $(-1, 21)$ is ramified exactly at $3, 7$. Both algebras split over $\mathbb{R}$, since their second entries are positive. Koenigsmann's trace-sum identity therefore gives

$$T_{-1,3} = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}, \qquad T_{-1,21} = \mathbb{Z}_{(3)} \cap \mathbb{Z}_{(7)}.$$

Their sum is $\mathbb{Z}_{(3)}$ by approximation, or directly by separating denominator factors at 2 and 7. This proves (13.1) using [4, Observation 5, Proposition 6, and Remark 8]. Thus membership in $\mathcal{R}_3$ is a fixed finite existential condition on a native rational variable.

13.2 Approximants whose indices are computed in the model

The prime 3 is inert and unramified in F, and E has good reduction there because $N_{F/\mathbb{Q}}(d) = 7$. Over $\mathbb{F}_9$ its reduction is $y^2 = x^3 + x$. The map $x \mapsto x^3 + x$ is the trace to $\mathbb{F}_3$, with three preimages of each value, and both nonzero elements of $\mathbb{F}_3$ are squares in $\mathbb{F}_9$. There are therefore 15 affine points and one point at infinity. Consequently

$$U = [16]P \neq O$$

belongs to the formal kernel at the unique place w above 3. Normalize $w(3) = 1$ and fix

$$\beta = N_{F/\mathbb{Q}}(z(U)) \in \mathbb{Q}^{\times}. \tag{13.2}$$

For $n \in \mathbb{Z}$ and $k \in \mathbb{N}$, define the points and native rational functions

$$B_k = [3^k]U = [16 \cdot 3^k]P, \qquad A_{n,k} = [n]B_k, \tag{13.3}$$

$$c_{n,k} = \frac{1}{2}\operatorname{Tr}_{F/\mathbb{Q}}\left(\frac{z(A_{n,k})}{z(B_k)}\right), \qquad b_k = \frac{N_{F/\mathbb{Q}}(z(B_k))}{\beta}. \tag{13.4}$$

When $n = 0$, use $z(A_{n,k}) = z(O) = 0$. The point B_k is always nonidentity and its parameter is nonzero.

Lemma 13.1. *For every $n \in \mathbb{Z}$ and $k \in \mathbb{N}$,*

$$v_3(c_{n,k} - n) \geq k + 1, \qquad v_3(b_k) = 2k. \tag{13.5}$$

Proof. The formal logarithm is an isometry on the formal kernel at 3, so

$$w(z(B_k)) = w(z(U)) + k \geq k + 1.$$

For each ordinary integer n, positive, negative, or zero, the multiplication series of the integral formal group has the form

$$[n](Z) = nZ + Z^2 F_n(Z), \qquad F_n(Z) \in \mathcal{O}_{F,w}[[Z]].$$

Evaluating at $z(B_k)$ gives

$$w\left(\frac{z(A_{n,k})}{z(B_k)} - n\right) \geq w(z(B_k)) \geq k + 1.$$

The trace divided by 2 does not decrease this lower bound, because 3 is unramified and 2 is a 3-adic unit. This proves the first assertion, including the case $n = 0$. The norm at the inert prime of residue degree two satisfies $v_3(N(a)) = 2w(a)$. The second assertion follows by subtracting the valuation of the fixed norm β. □

Theorem 13.2 (Uniform congruence to an integer label)**.** *There is a single finite existential formula* $\mathrm{Cong}_3(x, T, K)$ *such that*

$$\mathbb{Q} \models \mathrm{Cong}_3(x, nP, kP) \quad \Longleftrightarrow \quad v_3(x-n) \geq k \qquad (x \in \mathbb{Q},\ n \in \mathbb{Z},\ k \in \mathbb{N}). \tag{13.6}$$

The formula is false when its precision label is negative or either point parameter is not in G*.*

Proof. The relation on integer labels

$$\{(k, b) \in \mathbb{Z}^2 : k \geq 0,\ b = 16 \cdot 3^k\}$$

is computable. By Corollary 9.3, it has one fixed Diophantine definition on G^2. Use it to quantify $B = B_k$ from $K = kP$, and use $\mathrm{Mul}_G(T, B, A)$ to obtain $A = A_{n,k}$. Introduce c, b by the rational-function identities in (13.4), and require

$$(x-c)^2/b \in \mathcal{R}_3. \tag{13.7}$$

The denominator b is nonzero. The condition is expressed by an auxiliary variable $u \in \mathcal{R}_3$ and the equation $(x-c)^2 = bu$.

By Lemma 13.1, condition (13.7) is equivalent to

$$2v_3(x - c_{n,k}) - 2k \geq 0,$$

with the same interpretation when $x = c_{n,k}$. Since $v_3(c_{n,k} - n) \geq k+1$, this is equivalent to $v_3(x-n) \geq k$. All native operations in the construction are rational functions of a fixed number of point codes. Exponentiation occurs only in the fixed computable relation on the integer labels. There is no variable-degree formal-logarithm truncation, variable-length list of equations, or native scalar exponentiation in the formula. □

14 The height bound and exact scalar evaluation

For $n \in \mathbb{Z}$, put

$$m_i(n) = a_i n + b_i, \qquad R(n) = \prod_{i=1}^{4} |m_i(n)|, \qquad K(n) = HR(n)^c + |n| + 10. \tag{14.1}$$

The expression $K(n)$ is defined also when $R(n) = 0$. The ordinary branch below will exclude those cases. This is a fixed computable function on integer labels, so

$$\mathrm{Cut}(nP, kP) \quad \Longleftrightarrow \quad k = K(n)$$

is Diophantine by Corollary 9.3. In fact, apart from absolute values, it uses only a fixed number of ring operations and a fixed power.

Lemma 14.1 (Height bound from labelled contact support)**.** *Let* $n \in \mathbb{Z}$ *with* $m_i(n) \neq 0$ *for every* i*, and let* $x \in \mathbb{Q}$ *satisfy*

$$\mathrm{Supp}(a_i x + b_i, m_i(n)P) \qquad (1 \leq i \leq 4). \tag{14.2}$$

Then $x \notin \mathcal{C}$ *and*

$$h(x) \leq HR(n)^c. \tag{14.3}$$

Proof. Every native contact form $a_i x + b_i$ is nonzero by Definition 12.1, and x is finite, so $x \notin \mathcal{C}$. Fix a prime $q \notin S$ that does not divide $R(n)$. Proposition 12.2 implies that each raw valuation

$$v_q(a_i x + b_i)$$

is even. Set $\mu_q = \min\{v_q(x), 0\}$. If $v_q(x) \geq 0$, including $x = 0$, then $\mu_q = 0$. If $v_q(x) < 0$, every a_i is a q-adic unit and b_i is integral, so

$$v_q(a_i x + b_i) = v_q(x).$$

Thus μ_q is even in this case as well. The four finite normalized contact orders

$$v_q(a_i x + b_i) - \mu_q$$

and the contact order at infinity, namely $-\mu_q$, are all even. Consequently q does not occur in $M_S(x)$. Every prime that does occur divides $R(n)$, whence

$$M_S(x) \mid \mathrm{rad}(R(n)), \qquad M_S(x) \leq R(n).$$

Applying (10.2) proves (14.3). □

Let

$$\mathcal{E} = \{n \in \mathbb{Z} : a_i n + b_i = 0 \text{ for some } i\} = \mathcal{C} \cap \mathbb{Z}.$$

This is a fixed finite set of ordinary integers, determined by the four rational finite contact points.

Definition 14.2 (The scalar formula)**.** Require $T \in G$. Define $\mathrm{Eval}(x, T)$ to hold if either

$$\bigvee_{e \in \mathcal{E}} (x = e \ \wedge \ T = eP) \tag{14.4}$$

holds, or there exist $M_1, \ldots, M_4, K \in G$ such that

$$\begin{aligned} &M_i = [a_i]T + [b_i]P && (1 \leq i \leq 4), \\ &\mathrm{Supp}(a_i x + b_i, M_i) && (1 \leq i \leq 4), \\ &\mathrm{Cut}(T, K), \qquad \mathrm{Cong}_3(x, T, K). \end{aligned} \tag{14.5}$$

The empty exceptional disjunction is interpreted as false. Every multiplication $[a_i]$ or $[b_i]$ in the first line is by a fixed integer. The multiplication on variable labels is already included in the fixed predicate Supp.

Proof of Theorem 10.1. First let $x = n \in \mathbb{Z}$ and $T = nP$. If $n \in \mathcal{E}$, use (14.4). Otherwise put $M_i = m_i(n)P$ and $K = K(n)P$. Each support predicate is true by diagonal completeness, Proposition 12.3. The cutoff relation is true by its definition, and the congruence relation is true because $x - n = 0$. Thus every integer has witnesses, with its prescribed label, irrespective of its size or its prime divisors.

Conversely, fix $T = nP$ and suppose $\mathrm{Eval}(x, T)$ holds. The exceptional branch implies $x = n$ because P is nontorsion. In the ordinary branch, the point identities force $M_i = m_i(n)P$. The support predicates force every $m_i(n) \neq 0$, so $R(n) \geq 1$. Lemma 14.1 gives $h(x) \leq HR(n)^c$. The last two conditions force

$$v_3(x - n) \geq K(n). \tag{14.6}$$

Suppose for a contradiction that $x \neq n$. For a nonzero rational number y, a nonnegative valuation $v_3(y) \geq k$ implies $h(y) \geq k \log 3$, since its reduced numerator is divisible by 3^k. The usual height inequality for a difference gives

$$\begin{aligned} K(n) \log 3 &\leq h(x-n) \\ &\leq h(x) + h(n) + \log 2 \\ &\leq HR(n)^c + \log \max\{1, |n|\} + \log 2 \\ &< HR(n)^c + |n| + 10 = K(n). \end{aligned}$$

This is impossible because $\log 3 > 1$ and $K(n) > 0$. Therefore $x = n$.

All predicates in Definition 14.2 have finite existential definitions over $\mathbb{Q}$. The number of point variables, support certificates, and local predicates is fixed independently of x, n, and $K(n)$. Thus Eval is a single existential formula with the claimed graph. Existentially quantifying T defines precisely the literal subset $\mathbb{Z} \subseteq \mathbb{Q}$. □

Corollary 14.3 (One integer-coefficient polynomial)**.** *Under the hypotheses of Theorem 10.1, there are an integer $N \geq 1$ and a polynomial*

$$D(X, Y_1, \ldots, Y_N) \in \mathbb{Z}[X, Y_1, \ldots, Y_N]$$

such that

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^N \; D(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

Proof. Expand all the finitely many existential predicates and point charts in $\exists T$ Eval(X, T). Replace each inequation by an equation with a reciprocal witness. Move the finitely many existential quantifiers to the front, keeping denominator equations in the branches where they are used. Finite conjunctions of equations are combined by sums of squares over $\mathbb{Q}$, and finite disjunctions by products. Clear the fixed rational coefficient denominators. The construction is the same finite compilation justified in Section 8 and Appendix A. □

15 Consequences and checks of the new bridge

Why the earlier denominator obstruction is avoided. A fixed list of congruences about a fixed label admits rational perturbations with new denominators. The new formula does not rely on that list alone. Its four support predicates imply a height bound whose right-hand side is an ordinary integer function of the label. The fixed congruence formula then uses a precision label exceeding that bound. No inequality between a native rational height and an undefined native copy of a model integer appears in the syntax. The height comparison occurs only in the proof of soundness.

Why the weak height estimate suffices. The estimate used is exactly $h(x) \leq HM_S(x)^c$, for logarithmic height. There is no assumption of an estimate of the stronger form $h(x) \ll \log M_S(x)$. The exponent $K(n)$ is allowed to depend on the label, while its occurrence in the existential formula is uniform. This is the point that replaces the unproved stronger height hypothesis considered in the earlier attempt.

No loss of primes at the norm step. The support certificate forces J to be formal at all places above an odd-contact prime, by testing both rational coefficients of $1/x(J)$. This makes the valuation of the norm exactly $2v_q(m)$. Testing only one place above a split prime would not justify the same calculation.

Bad primes in honest witnesses. The support-soundness argument discards one fixed finite set S. Diagonal completeness, by contrast, works at every prime dividing the integer modulus, including primes in S. It uses arbitrarily deep local formal subgroups, not a good-reduction assertion at those bad primes. The integer-modulus localization lemma then supplies integer coefficient witnesses for every guard.

Uniformity and fixed constants. The constants H, c, the curve, the generator, and the finite contact configuration are chosen once. In particular, the exceptional prime set does not grow with $K(n)$. The support relation uses the exact formal-parameter valuation identity and only the precision-one rational localization certificate, while the 3-adic comparison increases the depth of the point B_k instead of increasing a logarithm truncation degree. Hence neither step requires the condition $q > K(n)+1$, and no precision-dependent factorial loss is hidden in H. Their numerical expansion is not supplied here. For each fixed choice the integer function $K(n)$ is computable, and the arithmetic inputs guarantee choices for which the correctness proof holds. The model represents the one fixed computable graph $b = 16 \cdot 3^k$; it does not run a native variable-length sequence of elliptic additions in the formula. The theorem establishes the existence of a single polynomial, not a printed expansion of its coefficient list.

No circular use of scalar evaluation. The support relation uses only multiplication on elliptic labels, which was constructed in Section 9 without the scalar bridge. The congruence relation uses computably enumerable relations on the same labels and the fixed local ring $\mathbb{Z}_{(3)}$. Only after both relations have been defined is the scalar evaluation formula constructed. Neither relation assumes that a native rational variable equals the integer represented by a point.

Corollary 15.1 (Rational Diophantine representations)**.** *Under the hypotheses of Theorem 10.1, every computably enumerable subset of $\mathbb{Q}^r$ is Diophantine over $\mathbb{Q}$.*

Proof. For a computably enumerable $A \subseteq \mathbb{Q}^r$, pull it back along the computable representation of rational tuples by integer numerators and nonzero integer denominators. The resulting computably enumerable integer relation has a Diophantine representation over $\mathbb{Z}$ by the Davis–Putnam–Robinson–Matiyasevich theorem. Use Corollary 14.3 to restrict the numerator, denominator, and auxiliary variables to the native integers, and impose the equations $x_i v_i = u_i$ with $v_i \neq 0$. This gives the required existential formula over $\mathbb{Q}$. □

Corollary 15.2 (Identity-parameter master polynomials)**.** *Under the same hypotheses, every computably enumerable relation $S \subseteq \mathbb{N} \times \mathbb{Q}^\times$ admits an integer-coefficient polynomial $M_S(X, Y, \mathbf{Z})$ such that, for $m \in \mathbb{N}$ and $b \in \mathbb{Q}^\times$,*

$$S(m,b) \Longrightarrow \exists \mathbf{z} \in (\mathbb{Q}^\times)^r \; M_S(m,b,\mathbf{z}) = 0,$$
$$\neg S(m,b) \Longrightarrow \nexists \mathbf{z} \in \mathbb{Q}^r \; M_S(m,b,\mathbf{z}) = 0.$$

Thus the external specialization and multiplicative parameter maps can both be the identity maps, $f(m) = m$ and $g(b) = b$.

Proof. View S as a computably enumerable subset of $\mathbb{Q}^2$, declared false outside the specified domain. Apply Corollary 15.1 and combine the defining equations into one integer-coefficient polynomial $P_S(X, Y, W_1, \ldots, W_e)$. Set $M_S(X, Y, \mathbf{U}, \mathbf{V}) = P_S(X, Y, U_1 - V_1, \ldots, U_e - V_e)$. Every rational value is a difference of two nonzero rationals, while any rational zero of the substituted polynomial produces a rational zero of P_S. This gives both required witness conditions. □

This is stronger than the earlier additive construction with $g = v_3$. Applications to Ramsey-theoretic complexity inherit the arithmetic dependencies and the provisional verification status of the present construction.

A Explicit existential encoding of elliptic points

For completeness we give finite rational encodings for all point operations used above. These observations also verify that the final small-index implications have no concealed universal quantifiers.

A.1 Arithmetic in the quadratic field

Represent $u = u_0 + u_1\sqrt{2} \in F$ by $(u_0, u_1) \in \mathbb{Q}^2$. The operation rules are

$$(u_0, u_1) + (v_0, v_1) = (u_0 + v_0, u_1 + v_1),$$

$$(u_0, u_1)(v_0, v_1) = (u_0v_0 + 2u_1v_1,\ u_0v_1 + u_1v_0), \qquad (u_0, u_1)^\sigma = (u_0, -u_1).$$

An F-equation is therefore two rational equations. The norm $u_0^2 - 2u_1^2$ is zero exactly when $u = 0$. Consequently nonzero conditions and the graph of an inverse are expressible by rational equations with existential witnesses. This interpretation never asks for a rational square root of 2.

A.2 Point codes and equality

A point code consists of $\epsilon \in \mathbb{Q}$ and $x, y \in F$, subject to

$$\mathrm{Pt}_E(\epsilon, x, y) : (\epsilon = 0 \ \wedge\ x = y = 0) \ \vee\ (\epsilon = 1 \ \wedge\ y^2 = x^3 - d^2x).$$

The first branch is the unique code for O. The second branch is the unique affine code for a finite point with those coordinates. Equality of valid codes is simply equality of their five rational components. If those components are $q_1, \dots, q_5$ and $q'_1, \dots, q'_5$, inequality is the rational inequation

$$\sum_{i=1}^{5} (q_i - q'_i)^2 \neq 0.$$

It becomes an existential equation by introducing its reciprocal. In particular $T \neq [s]P$ in (7.5) is an inequality with a fixed, computable point code, not the negation of a Diophantine image predicate. The same coding applies to E^σ, replacing d by d^σ.

A.3 Addition and fixed multiplication

For valid codes U, V, W, the condition $W = U + V$ is a finite disjunction of the following cases. If $U = O$, impose $W = V$; if $V = O$, impose $W = U$. Otherwise U, V are finite, and the remaining cases use their affine coordinates. Each of these remaining branches explicitly includes $\epsilon_U = \epsilon_V = 1$.

If $x_U = x_V$ and $y_U + y_V = 0$, impose $W = O$. This includes the doubling of a two-torsion point. If $x_U \neq x_V$, impose that W is finite and introduce $\lambda \in F$ satisfying

$$(x_V - x_U)\lambda = y_V - y_U, \qquad x_W = \lambda^2 - x_U - x_V, \qquad y_W = \lambda(x_U - x_W) - y_U.$$

Finally, if $x_U = x_V$, $y_U = y_V$, and $y_U \neq 0$, impose that W is finite and use

$$2y_U\lambda = 3x_U^2 - d^2, \qquad x_W = \lambda^2 - 2x_U, \qquad y_W = \lambda(x_U - x_W) - y_U.$$

These cases exhaust all possibilities on this nonsingular curve. They are equations and inequations over F and thus have finite existential rational encodings. Inversion changes y to $-y$ on the finite chart and fixes O. Repeated addition gives the graph of $[m]$ for the one fixed integer m. Accordingly membership in G is the existential assertion

$$\mathrm{Pt}_E(T) \ \wedge\ \exists U \quad \mathrm{Pt}_E(U) \ \wedge\ [m]U = T.$$

No instance of the negation of this assertion is needed.

A.4 Rational functions and the identity branch

On the identity branch define $z = 0$. On a finite branch where $y \neq 0$, impose $yz = -x$. On a finite branch where $x \neq 0$, define the reciprocal $t = 1/x$ by $xt = 1$. A quotient A/B in any certificate is represented by a variable D with $B \neq 0$ and $BD = A$. The root and logarithm denominators in the main construction are all explicitly nonzero. Auxiliary equations for a rational function remain inside the same disjunct as its domain restrictions; reciprocal-coordinate equations are not imposed on identity branches. Conjugation of point coordinates is the rational linear operation given above. Thus every displayed abbreviation expands to a finite existential Boolean combination of rational polynomial equations and inequations. The witnesses used for this expansion are not among the guarded localization coefficients unless explicitly listed in Definition 4.2.

B Dependency and quantifier audit

The following is a record of the logical interfaces of the conditional proof, not a set of additional assumptions.

$$\begin{aligned}
\text{Inputs P, E, H, R} &\Longrightarrow \text{the fiberwise formulas } \Theta_f,\\
\Theta_{X_1X_2-X_3} &\Longrightarrow \mathrm{Mul}_G,\\
\mathrm{Mul}_G + \text{DPRM over } \mathbb{Z} &\Longrightarrow \text{c.e. relations on labels},\\
\mathrm{Mul}_G + \Phi + \text{local elliptic theory} &\Longrightarrow \mathrm{Supp},\\
\text{label arithmetic} + \mathcal{R}_3 + \text{formal approximation} &\Longrightarrow \mathrm{Cong}_3,\\
\mathrm{Supp} + \mathrm{Cong}_3 + \text{Input H} &\Longrightarrow \mathrm{Eval}\,.
\end{aligned}$$

The predicates Cut and the label graph $b = 16 \cdot 3^k$ are fixed instances of c.e. relations on labels. Neither depends on a prior definition of scalar evaluation.

Two different uses of precision. In the first part, $k = c \deg f$ and the finite cutoff N_f are computed externally from one fixed polynomial f. Once $f = X_1X_2 - X_3$ is fixed, this defines one multiplication predicate of fixed arity. In the scalar bridge, the precision $K(n)$ is a variable integer label inside a fixed formula. It is realized through deeper formal points, not by substituting a variable into the exponent of a ring term or by writing a variable number of tests. The exceptional set and the height constants in that bridge are fixed independently of n.

Existence versus a printed coefficient expansion. The argument proves existence of a finite tuple of rational and integer parameters for which all formulas have the asserted meaning. Every such fixed tuple is legitimate finite data for an existential formula or a computable compiler. An unknown real-valued function of the input could not be used in this way; no such function occurs here. The manuscript specifies the fixed predicates and their finite compilation, but does not print the expanded integer polynomial or numerical values for all constants supplied by the inputs.

Edge cases. The identity point has one canonical code and the assigned value $z(O) = 0$. Reciprocal-coordinate equations are only imposed on the relevant nonidentity branches. The support relation is required only for nonzero integer labels and is diagonally complete for all such labels, including negative integers and ± 1. Nonintegral finite contact points cannot be admitted through the exceptional disjunction, which includes only $\mathcal{C} \cap \mathbb{Z}$. The case $n = 0$ is covered either by that finite disjunction or by the ordinary branch, where $A_{0,k} = O$.

Verification status and external dependencies

A further adversarial audit read the entire deduction, including the finite-certificate integer model, and checked the exact hypotheses of the four imported statements against the source manuscript. It identified no unresolved mathematical gap in the conditional deduction. The principal checks were the permitted denominators in the guards, simultaneous local witnesses, preservation of free point parameters, formal neighborhoods at bad and ramified primes, the norm weights at both split places, and fixed-formula variable precision. The specialization of the fixed local-ring definition was checked against the quaternionic trace-sum theorem. Separate exact-arithmetic checks verified the reduction-point count $\#E(\mathbb{F}_9) = 16$, the trace and norm formulas at precisions $0, 1$ on the source's explicit point for labels $-2, -1, 0, 1, 2$, and the elementary localization equations for positive and negative composite moduli. These finite checks are not substitutes for the general proofs and do not verify the existential predicates in their entirety. The four arithmetic inputs from [1] and their companion-paper dependencies remain assumed rather than independently verified. This manuscript is an AI-assisted conditional proof, not independent peer review or a machine-checked formalization.

References

[1] OpenAI. *Hilbert's tenth problem over the rational numbers*. Manuscript dated September 24, 2026; source statements checked October 11, 2026. The source is in the OpenAI mathematics repository: `https://github.com/openai/math/tree/main/preprints/Hilberts-tenth-problem-over-the-rational-numbers-September-24-2026`. The imported statements are the propositions labelled `par:pole-parity` and `ell:rank`, the theorem labelled `hei:height`, and the lemma labelled `pri:pattern`. Their statements and proofs are in the files `02-reduction.tex` through `06-primes.tex` in the `build/sections` directory. Section 7 explicitly distinguishes the source manuscript's Turing reduction from a many-one reduction.

[2] Martin Davis. *Hilbert's tenth problem is unsolvable*. The American Mathematical Monthly **80** (1973), no. 3, 233–269. `https://math.umd.edu/~mcl/Pubs/713/Diophantine.pdf`.

[3] Joseph H. Silverman. *The Arithmetic of Elliptic Curves*, second edition. Graduate Texts in Mathematics, vol. 106, Springer, New York, 2009. `https://doi.org/10.1007/978-0-387-09494-6`.

[4] Jochen Koenigsmann. *Defining $\mathbb{Z}$ in $\mathbb{Q}$*. Annals of Mathematics (2) **183** (2016), no. 1, 73–93. `https://arxiv.org/abs/1011.3424`. Proposition 6 and Remark 8 provide the quaternionic descriptions of the semilocal rings used in Section 13.

[5] Bjorn Poonen. *Using elliptic curves of rank one towards the undecidability of Hilbert's Tenth Problem over rings of algebraic integers*. In C. Fieker and D. Kohel (eds.), *Algorithmic Number Theory, ANTS-V*, Lecture Notes in Computer Science, vol. 2369, Springer, Berlin, 2002, pp. 33–42. `https://math.mit.edu/~poonen/papers/ants5.pdf`.